

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 321 — Past Questions & Materials
by Renance

Course: MTH 321
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — MTH 321 2017/2018

Level 300 • First Semester • PastQuestion

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(c) Prove that arbitrary union of subset of a metric : f

(ii) Noe Br MmGnleet ican open balk uustiny Cc oe

; QUESTION THREE

a) Define the terms (i) Cauchy Sequence {ii) Complete set a

{ii) PR oe Y meron sequence is Couch an

(b) Show that $x_n = \frac{1}{n}$, $\forall n > 1$ and $x_n \in [0,1]$ is Cauchy. &

(ii) Is the set $I = [0,1]$ complete? Justify.

(c) If $X = \mathbb{R}$ with the usual metric and (x_n) is given by $x_n = (2 + \frac{1}{n})$, Determine whether (x_n)

converges or not.

If it converges, find the limit and prove that the sequence

indeed converges to its limit.

QUESTION FOUR

a) Define continuity in terms of Open ball

b) State without proof, the continuity criterion in metric spaces and use it to determine the open sets of the following functions (i) $f(x) = x^2$ (ii) $f(x) = \exp(x)$ (iii) $f(x) = \sqrt{x}$

c) Prove using $\epsilon - \delta$ definition that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x, y) = x + y$ is continuous.

QUESTION FIVE

a) Define the following terms (i) Pointwise convergence of Sequence of RVF (ii) Uniform Convergence

of Sequence of RVF

b) Let for each $n \in \mathbb{N}$ and (X, d) be a metric space. Let $f_n: [0,1] \rightarrow \mathbb{R}$ be a function defined by

$f_n(x) = xe^{nx}$, $x \in [0, 1]$ and $n > 1$. Show that $\{f_n\}$ converges (i) Pointwise in (X, d) (ii) Uniformly in (X, d) .

c) Compute the integral of $f(x) = xe^{nx}$, $x \in [0,1]$. A

QUESTION SIX

a) What do you understand by (i) compactness and (ii) connectedness? Give 2 examples each.

b) Prove that a set is closed if and only if it contains all its limit points.

c) State without proof Heine-Borel theorem.

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ii) Show that the set of all rational numbers is neither open nor closed.

Rest wichest!

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